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Time Series Exploration and Preprocessing



Introduction to Time Series Data

What is Time Series Data?

- A time series is a collection of observations recorded over time.
- The order of the observations matters.
- Each observation has a date or time associated with it.

Examples :

- Daily temperatures
- Monthly airline passengers
- Stock prices
- Electricity usage



Dataset

Airline Passengers Dataset

- The AirPassengers dataset contains the monthly totals of international airline passengers from January 1949 to December 1960.
- Rows: 144 (one per month)

Columns:

- Month: Date (formatted as YYYY-MM)
- Passengers: Number of airline passengers that month (in thousands)



Basic Exploration

- `df.head()`, `df.tail()` etc
 - Verify the data loaded correctly.
 - Understand the structure.
 - Identify obvious errors.
-
- Get descriptive statistics
 - Check missing values

Preprocessing

Converting Dates

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>  
RangeIndex: 144 entries, 0 to 143  
Data columns (total 2 columns):  
#   Column      Non-Null Count  Dtype  
---  ---  
0   Month       144 non-null   object  
1   Passengers  144 non-null   int64  
dtypes: int64(1), object(1)  
memory usage: 2.4+ KB
```



```
df["Month"] = pd.to_datetime(df["Month"])
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>  
RangeIndex: 144 entries, 0 to 143  
Data columns (total 2 columns):  
#   Column      Non-Null Count  Dtype  
---  ---  
0   Month       144 non-null   datetime64[ns]  
1   Passengers  144 non-null   int64  
dtypes: datetime64[ns](1), int64(1)  
memory usage: 2.4 KB
```



Preprocessing

Set the Date as the Index

```
df.set_index("Month", inplace=True)
```

Before

- | Index | Month | Passengers |

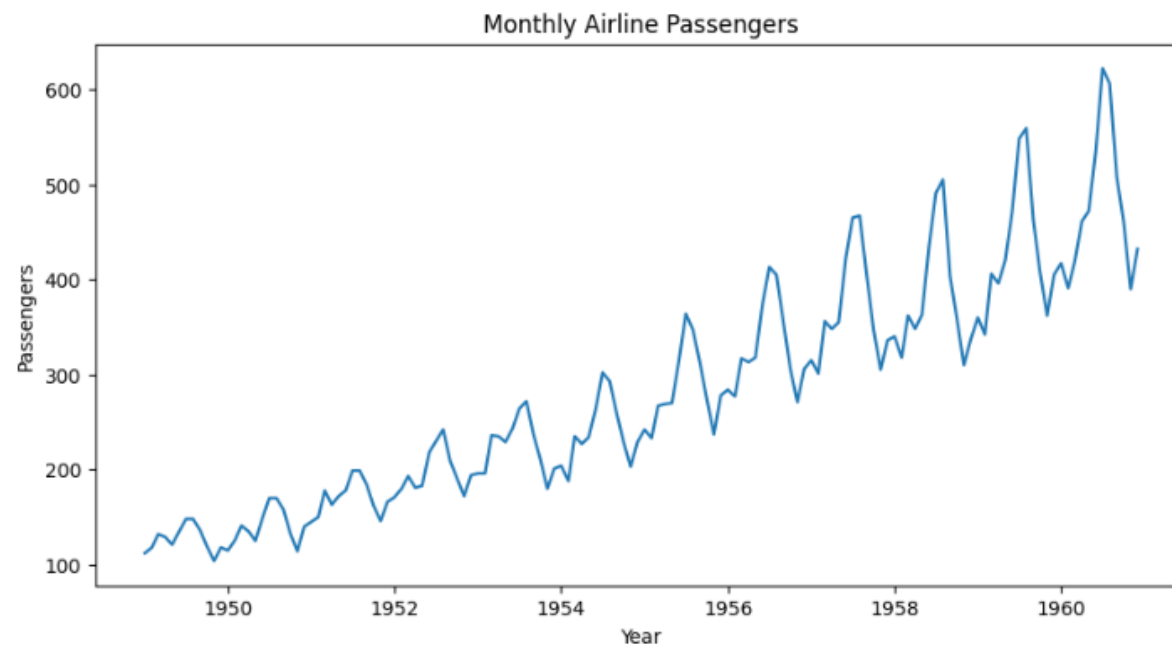
After

- | Month | Passengers |

Visualisation

```
import matplotlib.pyplot as plt
```

```
plt.figure(figsize=(10,5))  
plt.plot(df.index, df["Passengers"])  
plt.title("Monthly Airline Passengers")  
plt.xlabel("Year")  
plt.ylabel("Passengers")  
plt.show()
```





Resampling Time Series Data

Resampling means changing the frequency of a time series.

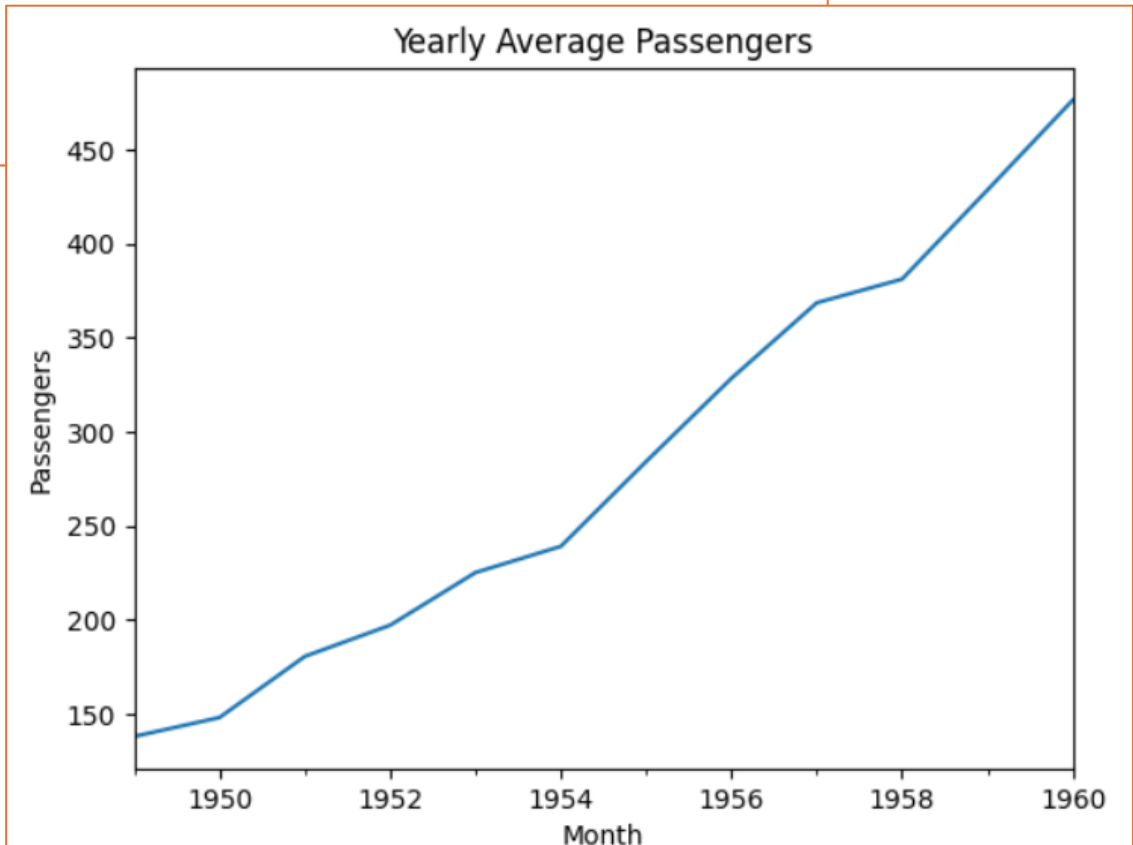
- Examples:
- Daily → Monthly
- Monthly → Yearly
- Hourly → Daily

This allows us to summarize or aggregate data over different time periods

Frequency	Code	Example
Daily	"D"	Every day
Weekly	"W"	Every week
Monthly	"M"	Every month
Quarterly	"Q"	Every 3 months
Yearly	"Y"	Every year

Resampling

```
# Resampling (Time-Based Grouping)  
# Yearly mean  
df['Passengers'].resample('Y').mean().plot(title='Yearly Average Passengers')  
  
plt.ylabel("Passengers")  
plt.show()
```





Rolling Statistics (Moving Window)

Rolling statistics calculate a statistic over a **moving window** of observations.

- Instead of calculating one average for the entire dataset, we calculate an average that moves through the data.

Example:

- Window 1: Jan Feb Mar
- Window 2: Feb Mar Apr
- Window 3: Mar Apr May

Each window produces a new average.



Rolling Statistics

Rolling mean example:

```
df["Rolling Mean"] = df["Passengers"].rolling(window=3).mean()  
  
df.head()
```

	Passengers	Rolling Mean
Month		
1949-01-01	112.0	NaN
1949-02-01	118.0	NaN
1949-03-01	132.0	120.666667
1949-04-01	129.0	126.333333
1949-05-01	121.0	127.333333

Rolling Statistics

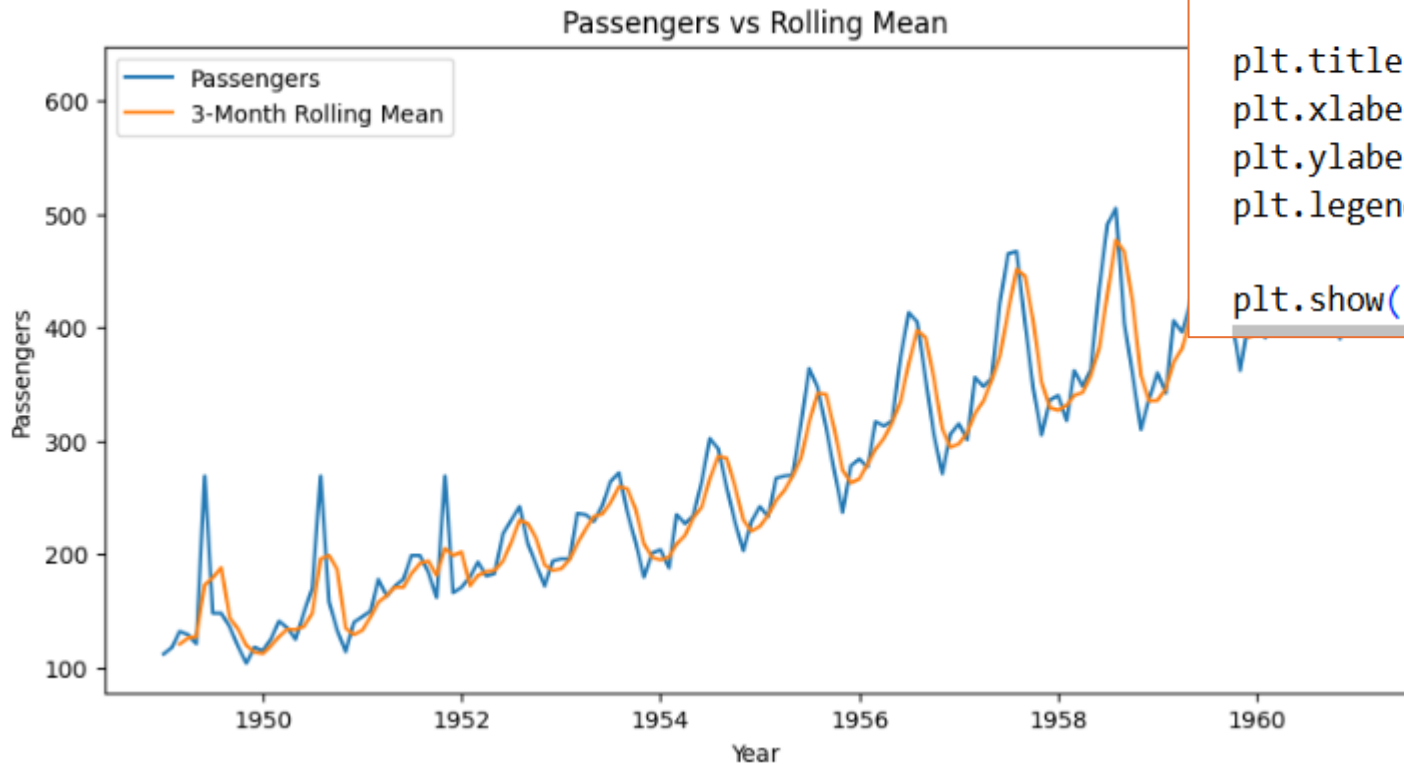
```
import matplotlib.pyplot as plt

plt.figure(figsize=(10,5))

plt.plot(df.index, df["Passengers"], label="Passengers")
plt.plot(df.index, df["Rolling Mean"], label="3-Month Rolling Mean")

plt.title("Passengers vs Rolling Mean")
plt.xlabel("Year")
plt.ylabel("Passengers")
plt.legend()

plt.show()
```





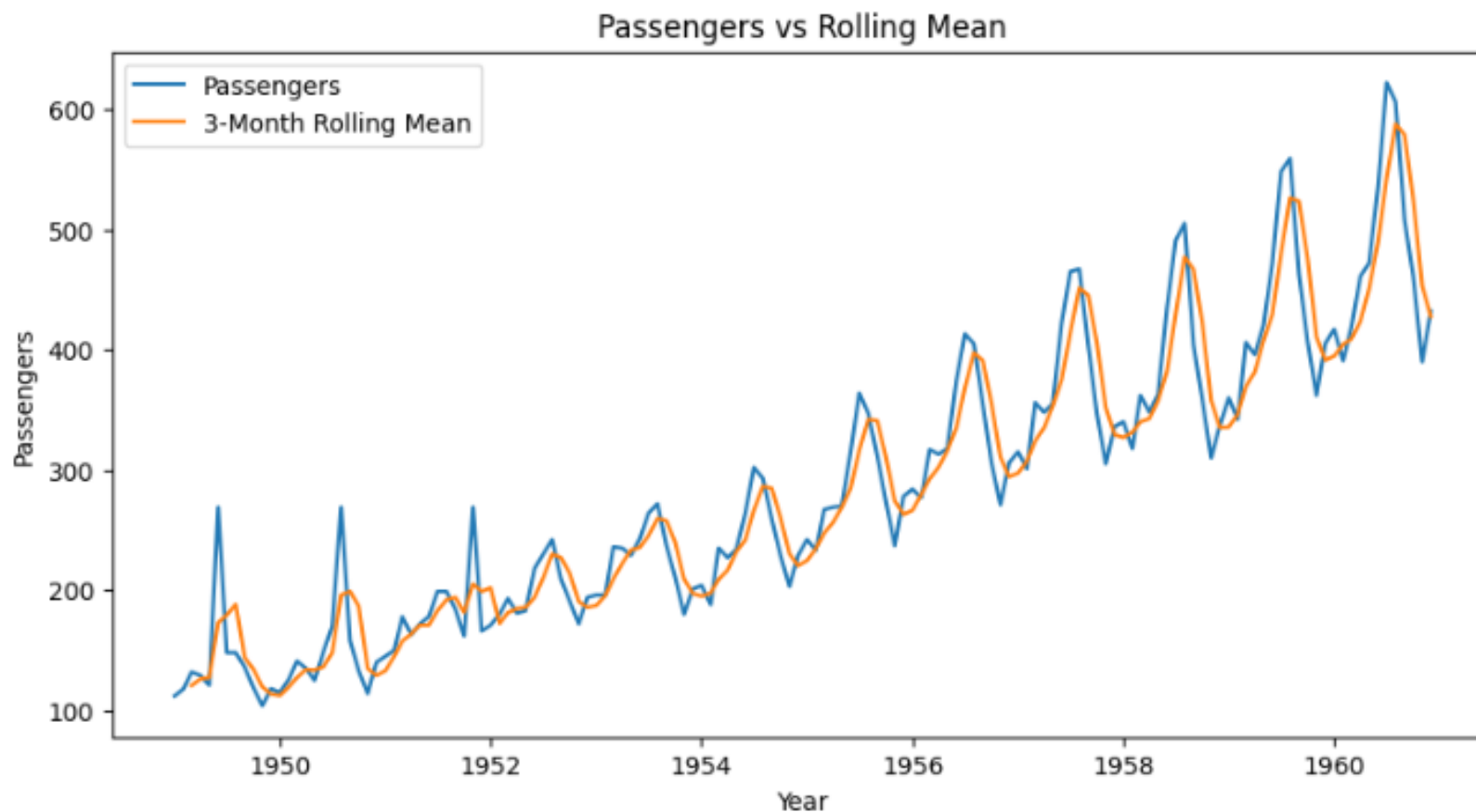
Rolling Statistics

Why Use Rolling Statistics?

- Smooths noisy data
- Highlights long-term trends
- Makes patterns easier to see
- Useful before forecasting



Comparing the Original Data and Rolling Mean





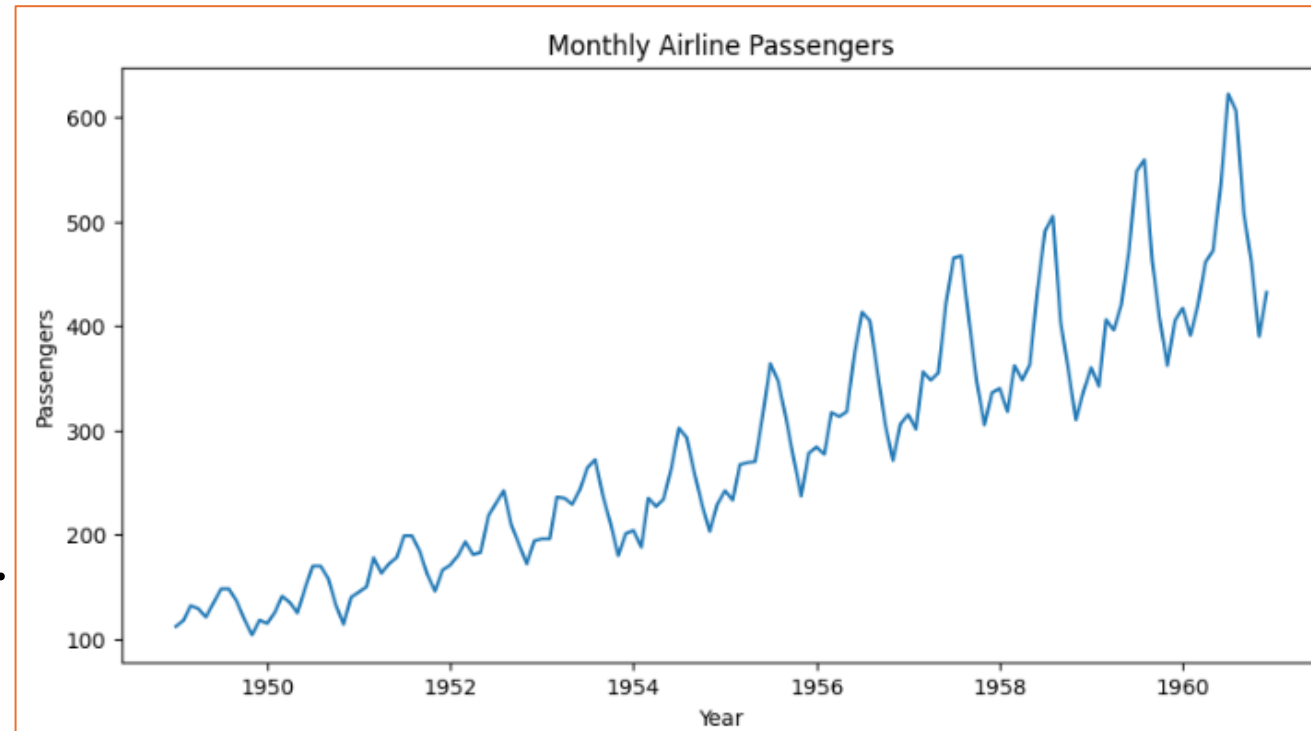
Trend & Seasonality

Trend

- A long-term increase or decrease in the data.
- Example
- Passenger numbers generally increase over time.

Seasonality

- Patterns that repeat at regular intervals.
- Example
- Passenger numbers rise during holiday periods every year





Key Takeaways

- Time series data has a date or time component.
- Always inspect the dataset before analysing it.
- Convert date columns to datetime.
- Set the date as the index.
- Visualise the data using a line plot.
- Look for trends and seasonality.



Activity

- Load the Airline Passengers dataset.
- Display the first five rows.
- Check the shape of the dataset.
- Use `info()` and identify the data types.
- Convert the **Month** column to datetime.
- Set **Month** as the index.
- Plot the passenger numbers.
- Describe the **trend** they observe.
- Describe the **seasonality** they observe.
- Create three missing values and use `SimpleImputer(strategy="median")` to fill them.

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